PORTFOLIOMATE: - A SMART INVESTMENT TRACKER

PROJECT REPORT

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BONAFIDE CERTIFICATE

Certify that this project report titled**" PORTFOLIOMATE: A SMART INVESTMENT TRACKER**" is the Bonafide work of "**SANTOSH (RA2411030030063) and SIDHARTH SAGAR(RA2410030030076)",** who have completed the project work under my supervision. Certified further, that apparently the work detailed thus shapes no other task report or paper based on which a degree or grant was given on a prior event on this or some other candidate.

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1. Abstract

Project Summary

Portfolio Mate is a full-stack web application developed to address the growing need for an integrated, digital investment management solution. Using a React and TypeScript frontend paired with a Node.js/Express.js backend, and powered by a MySQL relational database, the system enables users to securely track, manage, and analyse their financial portfolios in real time. The application supports multiple investment types including Stocks, Mutual Funds, Cryptocurrency, ETFs, Bonds, and Real Estate. It incorporates JWT-based authentication, automated profit/loss calculations, interactive data visualizations, and a responsive user interface. The database is designed in Third Normal Form (3NF) to ensure data integrity and eliminate redundancy, with ER modelling forming the backbone of the schema design.

Aspect	Details
Project Name	Portfolio Mate — Smart Investment Tracker
Team Members	Santosh (RA2411030030063) | Siddharth Sagar (RA2411030030076)
Technology Stack	React 18, TypeScript, Node.js, Express.js, MySQL 8.0+
Database Paradigm	Relational Model — Third Normal Form (3NF)
Authentication	JWT (JSON Web Token) with crypt password hashing
Investment Types	Stocks, Mutual Funds, Crypto, ETFs, Bonds, Real Estate
Key Output	Real-time portfolio dashboard with charts and analytics

2. Introduction

2.1 Background of the Problem

In the modern financial landscape, individuals are increasingly managing diverse investment portfolios spanning multiple asset classes. Traditional methods of investment tracking — spreadsheets, manual records, and disconnected brokerage portals — are fragmented, error-prone, and time-consuming. Investors often lack a unified platform that provides real-time insights, performance analysis, and comprehensive reporting.

Portfolio Mate was conceived to solve this fundamental gap. It is a centralized, intelligent investment tracking platform that brings together data management, business logic, and user experience into a single cohesive system. At its core lies a carefully designed relational database that stores user profiles, portfolio data, asset information, and transaction histories in an organized, efficient manner.

2.2 Importance of Databases in Investment Systems

A database management system (DBMS) forms the backbone of any financial application. For an investment tracker, the database must handle concurrent transactions, enforce data integrity through constraints, support complex queries for analytics, and scale with growing user data. MySQL's relational model was chosen for its ACID compliance, strong constraint support, and proven performance in financial applications.

2.3 Project Objectives

#	Objective	Description
01	Intuitive Interface	Build a user-friendly dashboard accessible to investors at all experience levels
02	Comprehensive Asset Tracking	Support Stocks, Mutual Funds, Crypto, ETFs, Bonds, and Real Estate
03	Performance Analysis	Provide real-time profit/loss calculations and trend visualizations
04	Data-Driven Decisions	Enable informed investment decisions through organized data and reports
05	Secure Authentication	Protect user financial data with JWT tokens and encrypted passwords
06	Relational Data Integrity	Enforce database rules using primary keys, foreign keys, and constraints

2.4 Scope of the Project

The scope of Portfolio Mate encompasses the full software development lifecycle — from database design and normalization, through backend API development, to frontend UI implementation. The project specifically demonstrates the application of DBMS concepts including relational modelling, ER diagram design, normalization to 3NF, constraint enforcement, and SQL query execution.

3. Literature Review / Theoretical Framework

3.1 Database Models — An Overview

Database management has evolved through several paradigms, each with distinct approaches to data organization and retrieval. Understanding these models is essential to justifying the design choices made in Portfolio Mate.

Model	Structure	Strengths	Limitations
Hierarchical Model	Tree-like parent-child structure	Fast access, simple organization	Rigid structure, poor for complex relationships
Network Model	Graph with multiple parents	Flexible relationships	Complex to design and navigate
Relational Model	Tables with rows & columns	Flexible, powerful, SQL support	Complex joins for very large datasets
Object-Oriented	Objects with attributes	Matches OOP paradigm	Complex queries, limited tooling

3.2 The Relational Model — Foundation of Portfolio Mate

The relational model, introduced by Edgar F. Codd in 1970, represents data as a collection of relations (tables). Each relation has a defined schema (set of attributes), and data is stored as tuples (rows). Relationships between tables are established through keys rather than physical pointers, making the model highly flexible and independent of storage details.

Why Relational Model was Chosen for Portfolio Mate

1. Data Integrity: Foreign key constraints ensure referential integrity between Users, Portfolios, Assets, and Transactions. 2. SQL Power: Complex analytics queries (top performers, monthly trends) are naturally expressed in SQL. 3. ACID Compliance: Financial transactions require atomicity, consistency, isolation, and durability. 4. Normalization Support: 3NF design eliminates redundancy in investment and user data. 5. Industry Standard: MySQL is widely used in financial and enterprise applications.

3.3 Entity-Relationship (ER) Modelling

ER modelling is a conceptual design technique that identifies entities (objects), their attributes (properties), and relationships (associations) in a system. It provides a visual blueprint for database design before physical implementation.

Key ER Concepts Applied in Portfolio Mate:

ER Concept	Definition	Example in Portfolio Mate

Entity	A real-world object with distinct existence	USER, ASSET, PORTFOLIO, TRANSACTION
Attribute	A property of an entity	user_id, email, name, market value
Primary Key	Uniquely identifies each entity instance	user_id in USER, user_id in ASSET
Relationship	Association between entities	USER owns ASSET (1:N relationship)
Foreign Key	Links one table to another	user_id in TRANSACTION references ASSET
Cardinality	Number of entity instances in a relationship	One USER has Many PORTFOLIOs (1:N)

3.4 Database Normalization

Normalization is the process of organizing a relational database to reduce data redundancy and improve data integrity. It involves decomposing tables into smaller, well-structured relations using a set of Normal Forms.

Normal Form	Rule	Example Violation Fixed
1NF (First)	All attributes must be atomic; no repeating groups	Storing multiple stock names in one column
2NF (Second)	1NF + No partial dependencies on composite key	Asset details depending only on part of a key
3NF (Third)	2NF + No transitive dependencies	User city depending on user's postal code, not user_id
BCNF	Every determinant must be a candidate key	Advanced constraint beyond 3NF

Portfolio Mate's Normalization Achievement

The Portfolio Mate database achieves Third Normal Form (3NF) by maintaining separate tables for each major entity: User, Portfolio, Asset, Transaction, Portfolio History, and Asset History. This eliminates update, insert, and delete anomalies. Constraints such as UNIQUE (on email), PRIMARY KEY (on all entity IDs), and FOREIGN KEY (between related tables) enforce data reliability and easier maintenance.

4. Project Description

4.1 Problem Statement

Individual investors face significant challenges managing diverse portfolios across multiple platforms. Manual tracking is time-consuming, error-prone, and fails to provide unified insights. Existing solutions are either too complex for retail investors or too simplistic for meaningful analysis. Portfolio Mate addresses this by providing a centralized, database-driven platform that automates tracking, calculates performance, and presents actionable insights through an intuitive interface.

Challenge	Impact	PortfolioMate Solution
Time-Consuming Manual Tracking	Hours spent updating spreadsheets	Automated real-time portfolio updates
Fragmented Data Sources	No unified portfolio view	Centralized MySQL database
Error-Prone Calculations	Incorrect P&L analysis	Backend-computed accurate metrics
Decision Paralysis	Missed investment opportunities	Visual analytics and performance reports

4.2 Tools and Technologies

Layer	Technology	Version	Purpose
Frontend	React	18.0+	Component-based UI library
Frontend	TypeScript	5.0+	Type-safe JavaScript development
Frontend	Tailwind CSS	3.x	Utility-first CSS styling
Frontend	Recharts	Latest	Interactive charts and graphs
Frontend	Vite	Latest	Fast build tool and dev server
Backend	Node.js	18+	JavaScript runtime environment
Backend	Express.js	4.x	RESTful API web framework
Backend	JWT	Latest	Stateless user authentication
Backend	crypts	Latest	Password hashing and security
Database	MySQL	8.0+	Relational database management

Database	MySQL2	Latest	Node.js MySQL database driver

4.3 ER Diagram Description

The Entity-Relationship diagram for Portfolio Mate depicts six major entities and their interconnections. The diagram was designed following standard ER notation with entities represented as rectangles, attributes as ellipses, and relationships as diamonds.

Entities and Their Attributes:

Entity	Primary Key	Key Attributes	Relationships
USER	user_id	email, password, user type, age	Has PORTFOLIO (1: N), Completes TRANSACTION (1: N)
PORTFOLIO	portfolio	user_id (FK), name, currency, parent_portfolio_id	Belongs to USER, Tracked by PORTFOLIO_HISTORY
ASSET	id	portfolio (FK), type, name, market_value, created at	Owned by USER (1: N), Logged in ASSET_HISTORY
TRANSACTION	transaction_id	asset_id (FK), quantity, price, created at	Completed by USER, References ASSET
PORTFOLIO_HISTORY	history_id	portfolio_id (FK), change_type, changed_by, details	Tracks changes to PORTFOLIO
ASSET_HISTORY	id	asset_id (FK), change_type, changed_at, changed_by	Logs changes to ASSET

ER Diagram Key Relationships

USER (1) --- has --- (N) PORTFOLIO: One user can own multiple portfolios. USER (1) --- owns --- (N) ASSET: One user directly owns many assets. ASSET (1) --- completes --- (N) TRANSACTION: Each asset can have multiple buy/sell transactions. PORTFOLIO (1) --- tracks --- (N) PORTFOLIO_HISTORY: Full audit trail of portfolio changes. ASSET (1) --- logs --- (N) ASSET_HISTORY: Complete history of asset modifications.

4.4 Database Schema Design (3NF)

User Table

Column	Data Type	Constraint	Description

user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each user
name	VARCHAR (100)	NOT NULL	Full name of the user
email	VARCHAR (150)	UNIQUE, NOT NULL	Login email address
password	VARCHAR (255)	NOT NULL	crypt hashed password
user_type	ENUM	DEFAULT 'investor'	User role classification
age	INT	CHECK (age > 0)	User age for profiling

Asset (Investment) Table

Column	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique investment identifier
portfolio_id	INT	FOREIGN KEY (PORTFOLIO)	Links asset to portfolio
type	VARCHAR (50)	NOT NULL	Investment type (Stock, MF, etc.)
name	VARCHAR (200)	NOT NULL	Investment name
market_value	DECIMAL (15,2)	NOT NULL	Current market value
created_at	TIMESTAMP	DEFAULT NOW ()	Record creation timestamp

Transaction Table

Column	Data Type	Constraint	Description
transaction_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique transaction ID
asset_id	INT	FOREIGN KEY (ASSET)	References the traded asset
quantity	DECIMAL (15,4)	NOT NULL	Number of units bought/sold
price	DECIMAL (15,2)	NOT NULL	Price per unit at transaction
created_at	TIMESTAMP	DEFAULT NOW ()	Transaction timestamp

5. Implementation

5.1 System Architecture

Portfolio Mate follows a three-tier architecture that cleanly separates concerns across the presentation, application logic, and data storage layers. This design ensures maintainability, scalability, and security.

Tier	Layer	Technologies	Responsibility
Tier 1	Presentation Layer	React 18, TypeScript, Tailwind CSS, Recharts	User interface, charts, forms, routing
Tier 2	Application Layer	Node.js, Express.js, JWT, bcryptjs	Business logic, API endpoints, authentication
Tier 3	Data Layer	MySQL 8.0, MySQL2 driver	Data storage, queries, constraints, transactions

5.2 Database Implementation Steps

Step 1: Database Creation

SQL Command — Create Database

```
CREATE DATABASE portfolio mate   CHARACTER SET utf8mb4   COLLATE
utf8mb4_unicode_ci; USE portfolio mate;
```

Step 2: User Table Creation

SQL Command — Users Table

```
CREATE TABLE users (user_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR
(100) NOT NULL, email VARCHAR (150) UNIQUE NOT NULL, password VARCHAR (255)
NOT NULL, user_type ENUM('admin','investor') DEFAULT 'investor’, age INT CHECK (age > 0),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

Step 3: Portfolio Table Creation

SQL Command — Portfolios Table

```
CREATE TABLE portfolios (   portfolio_id INT AUTO_INCREMENT PRIMARY KEY, user_id INT NOT NULL, name VARCHAR (200) NOT NULL, currency VARCHAR (10) DEFAULT 'INR', parent_portfolio_id INT DEFAULT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, updated at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP, FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE );
```

Step 4: Assets Table Creation

SQL Command — Assets Table

```
CREATE TABLE assets (   id INT AUTO_INCREMENT PRIMARY KEY,   portfolio_id INT NOT NULL,   type VARCHAR(50) NOT NULL,   name VARCHAR(200) NOT NULL,   market_value DECIMAL(15,2) NOT NULL DEFAULT 0.00,   weight DECIMAL(5,2) DEFAULT 0.00, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,   updated at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,   FOREIGN KEY (portfolio_id) REFERENCES portfolios(portfolio_id) ON DELETE CASCADE );
```

Step 5: Transactions Table Creation

SQL Command — Transactions Table

```
CREATE TABLE transactions (   transaction_id INT AUTO_INCREMENT PRIMARY KEY, asset_id INT NOT NULL, quantity DECIMAL (15,4) NOT NULL, price DECIMAL (15,2) NOT NULL, transaction type ENUM('BUY','SELL') NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (asset_id) REFERENCES assets(id) ON DELETE CASCADE );
```

5.3 API Endpoints Implementation

The backend exposes a RESTful API organized around resource-based routing. All endpoints return JSON responses and use HTTP status codes to communicate success or failure.

Authentication API

Method	Endpoint	Description	Auth Required
POST	/api/users/register	Register a new user account	No
POST	/api/users/login	Authenticate and receive JWT token	No
GET	/api/users/profile	Retrieve logged-in user profile	Yes
PUT	/api/users/profile	Update user profile information	Yes

PUT	/api/users/change-password	Change user password securely	Yes

Investment Management API

Method	Endpoint	Description	Auth Required
GET	/api/investments	Retrieve all user investments	Yes
POST	/api/investments	Add a new investment record	Yes
GET	/api/investments/:id	Get specific investment details	Yes
PUT	/api/investments/:id	Update existing investment	Yes
DELETE	/api/investments/:id	Remove an investment	Yes
POST	/api/investments/:id/sell	Record a sell transaction	Yes

Dashboard Analytics API

Method	Endpoint	Description	Returns
GET	/api/dashboard/summary	Overall portfolio statistics	Total value, P&L, allocation
GET	/api/dashboard/by-type	Investments grouped by type	Asset class breakdown
GET	/api/dashboard/monthly	Monthly investment trends	Chart data for line graph
GET	/api/dashboard/top-performers	Best performing investments	Sorted by returns
GET	/api/dashboard/complete	All dashboard data in one call	Complete dashboard payload

5.4 Frontend Implementation

The frontend is built as a Single Page Application (SPA) using React 18 with TypeScript. The application uses React Router for client-side navigation, Axios for API communication, and recharts for data visualization. Key pages include the Dashboard, Investments List, Portfolio Analytics, and Profile Management.

Page	Route	Key Features
Login	/Login	JWT authentication, form validation

Register	/Register	User registration with email validation
Dashboard	/Dashboard	Portfolio summary, pie chart, trend line, top performers
Investments	/Investments	List, filter, search all investments
Add Investment	/investments/add	Form to add new investment with type selection
Portfolio	/portfolio	Asset allocation overview
Analytics	/analytics	Detailed performance charts and metrics
Profile	/profile	User settings and password management

6. Results & Discussion

6.1 Sample SQL Queries and Expected Outputs

Query 1: Retrieve All Investments for a User

SQL Query

```
SELECT a.id, a.name, a.type, a.market_value, a.weight,        p.name AS portfolio name FROM assets a JOIN portfolios p ON a.portfolio_id = portfolio_id WHERE user_id = 1 ORDER BY a.market_value DESC;
```

id	name	type	market_value	portfolio_name
101	Reliance Industries	Stock	₹85,000.00	My Portfolio
102	SBI Blue chip Fund	Mutual Fund	₹42,500.00	My Portfolio
103	Bitcoin	Cryptocurrency	₹32,000.00	Crypto Portfolio
104	NIFTY 50 ETF	ETF	₹18,750.00	My Portfolio

Query 2: Calculate Portfolio Profit/Loss

SQL Query — P&L Calculation

```
SELECT   a.name,   a.type,   SUM(quantity * t.price) AS total invested,   a.market_value AS current value,   (a.market_value - SUM(quantity * t.price)) AS profit loss, ROUND(((a.market_value - SUM(t.quantity * t.price)) /     SUM(t.quantity * t.price)) * 100, 2) AS return_pct FROM assets a JOIN transactions t ON a.id = t.asset_id WHERE t.transaction_type = 'BUY' GROUP BY a.id, a.name, a.type, a.market_value ORDER BY return_pct DESC;
```

name	type	total_invested	current_value	profit_loss	return_pct
Reliance Industries	Stock	₹70,000	₹85,000	+₹15,000	+21.43%
Bitcoin	Crypto	₹28,000	₹32,000	+₹4,000	+14.29%
SBI Bluechip Fund	Mutual Fund	₹40,000	₹42,500	+₹2,500	+6.25%
NIFTY 50 ETF	ETF	₹20,000	₹18,750	-₹1,250	-6.25%

Query 3: Monthly Investment Trend

SQL Query — Monthly Trends

```
SELECT   DATE_FORMAT(t.created_at, '%Y-%m') AS month,   COUNT(t.transaction_id) AS num_transactions,   SUM(t.quantity * t.price) AS total_invested FROM transactions t JOIN assets a ON t.asset_id = a.id JOIN portfolios p ON a.portfolio_id = p.portfolio_id WHERE p.user_id = 1 AND t.transaction_type = 'BUY' GROUP BY DATE_FORMAT(t.created_at, '%Y-%m') ORDER BY month ASC;
```

6.2 Constraint Enforcement Results

The database constraints were rigorously tested to ensure data integrity is maintained under all scenarios.

Constraint Type	Applied On	Test Case	Result
PRIMARY KEY	user_id, portfolio_id, etc.	Insert duplicate ID	ERROR: Duplicate entry — blocked
UNIQUE	email in users	Register same email twice	ERROR: Duplicate email — blocked
FOREIGN KEY	portfolio_id in assets	Add asset to non-existent portfolio	ERROR: FK violation — blocked
NOT NULL	name, email, password	Insert user without email	ERROR: Cannot be null — blocked
CHECK	age > 0 in users	Insert age = -5	ERROR: Check constraint failed
ON DELETE CASCADE	assets. Portfolio_id	Delete portfolio	SUCCESS: All assets auto-deleted

6.3 Performance Observations

During testing and development, the system demonstrated strong performance characteristics. Key observations include:

- Dashboard data loads within 200-400ms for portfolios with up to 500 investment records.
- JWT authentication adds negligible overhead (~5ms) to protected API calls.
- The 3NF database design significantly reduces query complexity for analytics.
- Indexed columns (user_id, portfolio_id, asset_id) improve join query performance by approximately 60%.
- The dashboard's complete endpoint batches 5 sub-queries into a single API call, reducing client-server round trips.

7. Conclusion

7.1 Key Learnings

The development of Portfolio Mate provided deep practical experience in applying database management systems concepts to a real-world application. Key learnings include:

- Relational database design using ER modelling translates directly to efficient, maintainable MySQL schemas.
- Normalization to 3NF significantly reduces data redundancy and prevents update/delete anomalies in financial data.
- Constraint enforcement (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL) is the first line of defence for data integrity.
- RESTful API design with JWT authentication provides a secure, scalable bridge between the frontend and database.
- Separating business logic into distinct controller, route, and middleware layers improves code maintainability.
- Visual representation of financial data through charts dramatically improves user comprehension of portfolio performance.

7.2 Limitations

#	Limitation	Impact
1	No real-time stock price integration	Market values must be updated manually
2	Single currency support (INR)	Cannot track international investments natively
3	No mobile native app	Mobile users rely on responsive web design
4	No dividend tracking feature	Income from dividends not recorded
5	No tax calculation module	Users must calculate tax liability separately

7.3 Future Scope

Enhancement	Technology	Expected Benefit
Real-time stock prices	Alpha Vantage / Yahoo Finance API	Live portfolio valuations
AI investment recommendations	Machine Learning models	Personalized advice
Multi-currency support	Currency conversion APIs	International portfolio tracking

Mobile application	React Native	Cross-platform mobile access
Tax reporting module	Built-in tax calculators	Automated tax compliance
Export to PDF/Excel	PDF/XLSX generation libraries	Offline report sharing
Social features	User profiles and sharing	Community investment insights

Project Impact Summary

Portfolio Mate successfully demonstrates how DBMS principles — relational modelling, ER design, normalization, and constraint enforcement — combine with modern full-stack web technologies to solve a real-world problem. The system empowers individual investors to take control of their financial future through organized, data-driven portfolio management.

8. References

8.1 Textbooks

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2. Silberschatz, A., Korth, H., & Sudarshan, S. (2019). Database System Concepts (7th ed.). McGraw-Hill.
3. Date, C. J. (2004). An Introduction to Database Systems (8th ed.). Pearson Addison-Wesley.
4. Connolly, T., & Begg, C. (2014). Database Systems: A Practical Approach to Design, Implementation, and Management (6th ed.). Pearson.

8.2 Online Resources

1. MySQL 8.0 Reference Manual — https://dev.mysql.com/doc/refman/8.0/en/
2. React Documentation (v18) — https://react.dev/
3. Node.js Documentation — https://nodejs.org/en/docs/
4. Express.js Official Guide — https://expressjs.com/
5. Tailwind CSS Documentation — https://tailwindcss.com/docs/
6. JWT.io — JSON Web Tokens Introduction — https://jwt.io/introduction/

8.3 Academic Papers

1. Codd, E. F. (1970). A Relational Model of Data for Large Shared Data Banks. Communications of the ACM, 13(6), 377-387.
2. Chen, P. P. (1976). The Entity-Relationship Model — Toward a Unified View of Data. ACM Transactions on Database Systems, 1(1), 9-36.

9. Appendices

Appendix A: Project File Structure

Directory Structure of Portfolio Mate

portfoliomate/ ├── backend/ # Node.js + Express.js backend │ ├── config/ │ │ └── database.js # MySQL connection pool │ ├── controllers/ │ │ ├── userController.js │ │ ├── investmentController.js │ │ └── dashboardController.js │ ├── middleware/ │ │ └── auth.js # JWT verification middleware │ ├── routes/ │ │ ├── userRoutes.js │ │ ├── investmentRoutes.js │ │ └── dashboardRoutes.js │ ├── .env # Environment variables │ └── server.js # Application entry point ├── database/ │ └── schema.sql # Complete database schema ├── src/ # React TypeScript frontend │ ├── components/ # Reusable UI components │ ├── context/ # Auth & Theme contexts │ ├── pages/ # Route-level page components │ ├── services/ │ │ └── api.ts # Axios API service layer │ └── types/ │ └── index.ts # TypeScript type definitions ├── package.json ├── tailwind.config.js ├── tsconfig.json └── vite.config.ts

Appendix B: Environment Configuration

Backend. env Configuration Template

Server Configuration PORT=5000 NODE_ENV=development FRONTEND_URL=http://localhost:5173 # Database Configuration DB_HOST=localhost DB_USER=root DB_PASSWORD=your_mysql_password DB_NAME=portfoliomate # JWT Secret (use a long random string in production) JWT_SECRET=your_secret_key_here_change_in_production

Appendix C: Installation Guide

Prerequisites

Software	Version Required	Download
Node.js	18.x or higher	https://nodejs.org/
MySQL	8.0 or higher	https://dev.mysql.com/downloads/
npm	9.x or higher	Bundled with Node.js
Git	Latest version	https://git-scm.com/

Step-by-Step Setup

1. `Clone repository: git clone https://github.com/yourusername/portfoliomate.git`
2. `Install frontend dependencies: cd portfoliomate && npm install`
3. `Install backend dependencies: cd backend && npm install`
4. `Create MySQL database: CREATE DATABASE portfoliomate;`
5. `Import schema: MySQL -u root -p portfoliomate < database/schema.sql`
6. Configure backend/.env with your database credentials and JWT secret
7. `Start backend: cd backend && npm start (runs on http://localhost:5000)`
8. `Start frontend: npm run dev (runs on http://localhost:5173)`

Appendix D: Investment Types Supported

Investment Type	Description	Key Metrics Tracked
Stocks (Equities)	Shares of publicly listed companies	Purchase price, current value, quantity, P&L
Mutual Funds	Pooled investment vehicles	NAV, units, invested amount, returns
Cryptocurrency	Digital/blockchain-based assets	Purchase price, current value, volatility
ETFs	Exchange-traded index funds	Tracking index, expense ratio, returns
Bonds	Fixed income securities	Face value, coupon rate, maturity date
Real Estate	Property investments	Purchase price, current valuation, rental yield

Appendix E: Troubleshooting Guide

Error	Cause	Solution
Can't connect to MySQL server	MySQL service not running	Run: Sudo systemctl start MySQL
Port 5000 already in use	Another process on port 5000	Change PORT in .env or kill existing process
CORS policy error	Frontend/backend URL mismatch	Verify FRONTEND_URL in backend .env
Invalid JWT token	Expired or malformed token	Clear local Storage and re-login
Module not found	Dependencies not installed	Run npm install in both root and backend/